

David Oluwatimilehin “Timi” Folayan

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EDUCATION

Rose-Hulman Institute of Technology

Terre Haute, IN

Bachelor of Science in Software Engineering; Minor in Geography

Expected 2027

- Relevant Coursework: Object-Oriented Development, Operating Systems, Web Development, Data Structures & Algorithms, Digital Systems, Databases, Formal Methods, Practical Security, Software Design

TECHNICAL SKILLS

Languages: Python, Java, SQL, C#, TypeScript, JavaScript, HTML/CSS, C, C++

Tools & Platforms: Git, Docker, Kubernetes, FastAPI, PostgreSQL, Microsoft SQL Server, MLflow, scikit-learn, MCP, Supabase, pytest, NI-DAQmx, Cursor, VS Code, IntelliJ, Eclipse, React

EXPERIENCE

Emerson-NI (National Instruments)

Austin, TX

Sales Development Engineering Intern

Summer 2026

- Developed opportunity pipeline from inbound demand and proposed tailored hardware/software solutions to prospective customers, generating over \$1,600,000 in individual opportunity dollars
- Led live demonstrations of the MioDAQ platform at a company-wide Tech Expo, translating complex technical concepts into customer value to drive new business and relationships
- Built and documented internal projects showcasing NI's hardware/software capabilities, supporting community outreach and increasing NI's visibility
- Used LinkedIn Sales Navigator to build account intelligence and qualify prospects, supporting sales engineers in identifying new opportunities

RHV – Gas Valve Calibration System

Terre Haute, IN

Software Engineer Intern

Summer 2025

- Sole software engineer on a multidisciplinary team of electrical, computer, and mechanical engineers; built a multithreaded C# WinForms application as the central control interface for a six-unit calibration system
- Implemented Modbus TCP/IP and serial communication to coordinate PLCs and programmable power supplies, synchronizing thermal, pressure, and flow testing in real time across all six units
- Enabled high-throughput calibration of over 1,000,000 gas valves with minimal operator intervention

Rose-Hulman

Terre Haute, IN

Teaching Assistant, Formal Methods

Spring 2026

- Supported undergraduates in mastering logic, specifications, and program correctness; graded assignments and gave detailed feedback on formal proofs and model-based designs

PROJECTS

DAQ MCP Server | *Python, FastMCP, nidaqmx, Starlette, pytest*

- Built an MCP server so AI coding clients (Cursor, cloud agents) can discover, read, and write NI DAQ channels through structured tools instead of ad-hoc scripts
- Implemented a 5-layer safety model (writes off by default, channel allowlists, direction enforcement, analog-output clamping, digital write read-back) shared by agent tools and a live browser dashboard
- Abstracted hardware behind a pure-Python simulator plus a real NI-DAQmx backend so the same tool surface runs with zero NI hardware, covered by pytest in GitHub Actions

NBA Player Predictor | *Python, scikit-learn, FastAPI, PostgreSQL, MLflow, Docker, Kubernetes*

- Built a matchup-aware RandomForestRegressor on player/team data in PostgreSQL to predict next-game points against a specific opponent, replacing opaque IDs with scoring averages and opponent points allowed
- Tracked 10 MLflow runs varying tree depth and test split, promoting the lowest-MAE model to the Model Registry under a production alias
- Deployed via containerized FastAPI with CI/CD through GitHub Actions to GKE, automating QA releases and gating production for reliability

LinkedLife | *TypeScript, React, Vite, Zustand, Vitest, Vercel*

- Designed a deterministic, seeded game engine across 85 TypeScript modules (16K lines) so a satirical corporate life-sim is fully playable with zero API calls
- Split AI so a local classifier owns all game consequences and an LLM, if present, only writes flavor text; 271 automated tests include headless bot playthroughs over thousands of turns
- Authored branching dialogue for 8 personalities with tests walking 4,000+ conversation paths; 9 versioned save-schema migrations keep saves loading from the first build

LEADERSHIP & INVOLVEMENT

Vice President, African Student Union (2024–Present) · **Public Relations Chair**, National Society of Black Engineers (2024–Present)

Member, Linux Users Group, Computer Security Club, ColorStack (2023–Present)